

MAKE UP EXAMINATIONS - APRIL 2024

Program	: B.E. - Common to all Programs	Semester	: V
Course Name	: Research Methodology and Intellectual Property Rights	Max. Marks	: 100
Course Code	: AL58	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Suitably assume any missing data.
- Use of statistical tables are permitted.

UNIT - I

- Discuss the classification of research. CO1 (10)
 - What are the various types of plagiarism? Explain. CO1 (10)
- Write a short note on literature review. Describe the utility of Google scholar in the process of literature review. CO1 (10)
 - Explain the basic principles of research and its objectives. CO1 (10)

UNIT - II

- What is research design? Why is it important? Explain with examples. CO2 (08)
 - Explain the following with suitable examples: CO2 (06)
 - Dependent and independent variables
 - Extraneous variable
 - Confounded relationship.
 - What is research hypothesis? What is its purpose? CO2 (06)
- What is experimental design? Explain its principles. CO2 (10)
 - What is randomized block design? State its advantages and disadvantages. CO2 (10)

UNIT - III

- Elaborate the steps involved in Secondary Data Collection. CO3 (10)
 - What is observation? Discuss the different types of observation methods in research and add a note on their strengths and weaknesses. CO3 (10)
- Illustrate process of complex random sampling with suitable examples. Differentiate the cluster sampling from stratified sampling. CO3 (10)
 - What is telephonic interview? List out its advantages and disadvantages. CO3 (04)
 - According to the Indian Agricultural Department, in 2019 the average farm size in the state of Haryana was 3.5 km². Since the decades there has been a trend of farm size increasing due to large agribusinesses, an agribusiness analyst wishes to test if the current (2023) farm size is larger than it was in 2017. Frame the null and alternate hypothesis. What could be the possible type I and type II error in the decision? CO3 (06)

UNIT- IV

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| 7. | a) | Briefly explain the procedure for obtaining the patent in India. | CO4 | (10) |
| | b) | Describe all the Non-patentable inventions. | CO4 | (10) |
| 8. | a) | Explain principles underlying the patent law in India. | CO4 | (10) |
| | b) | Briefly explain TRIPS Agreement. | CO4 | (05) |
| | c) | Discuss Patent Cooperation Treaty (PCT). | CO4 | (05) |

UNIT- V

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| 9. | a) | Explain the registration procedure for trademarks. | CO5 | (10) |
| | b) | Explain the remedies against the infringement of copyrights. | CO5 | (10) |
| 10. | a) | Explain the essential requirements for registration of design. | CO5 | (10) |
| | b) | Describe different ways for protection of trademarks. | CO5 | (10) |
